Dereje D. Jima

MSc, PhD | Senior Bioinformatics Scientist

Human Health and Disease | Epidemiologic Research | Multi-omics

Cary, NC, USA | (919) 271-8057 | ddjima2014@gmail.com | www.ddjima.com
LinkedIn: dereje-jima-msc-phd-5395ab8 | GitHub: ddjima
ORCID: 0000-0002-7784-1612 | ResearchGate: Dereje-Jima

EDUCATION

PhD, Bioinformatics
North Carolina State University | 2022 -- 2026

MS in Bioinformatics
Wageningen University | 2003 -- 2005

BS in Plant Science
Haramaya University | 1996 -- 2000

PROFESSIONAL SUMMARY

- Senior bioinformatics scientist with 20+ years of experience in genomics, epigenomics, and multi-omics. Uses R/Bioconductor as a primary analytical environment and Python as needed for high-throughput analysis, automation, and reproducible workflow development.
- Demonstrated impact in integrating complex biological data to generate actionable scientific insights across cancer genomics, immunology, infectious disease, and environmental health studies, with strong emphasis on study design, analytical rigor, and scientific interpretation.
- Proven leader in cross-functional bioinformatics support, partnering with 50+ investigators across three universities to design sequencing studies, develop analytical workflows, produce high-quality reports, and translate data into defensible scientific conclusions.

SELECTED IMPACT

- **76 peer-reviewed publications:** Extensive publication record across human health and disease, epidemiologic research, genomics, epigenomics, and biomedical data science.
- **Cross-institution collaboration:** Bioinformatics support and consultation for investigators across multiple universities and research centers.
- **Broad omics expertise:** Sequencing, methylation, expression, and integrative analysis across diverse study designs.

PROFESSIONAL EXPERIENCE

Senior Research Scholar
North Carolina State University | Raleigh, NC
November 2023 -- Present

- Promoted from Research Scholar in recognition of sustained leadership in bioinformatics, multi-omics analysis, and cross-institutional scientific collaboration.
- Lead bioinformatics strategy as CHHE Liaison, partnering with 50+ investigators across three universities to support sequencing study design, multi-omics analysis, and interpretation of high-dimensional datasets.
- Serve as co-investigator on multiple NIH-funded projects, including R01, R21, and CHHE Center grants, contributing to proposal development, analytical planning, and leadership of bioinformatics components in funded research.
- Produce and review analytical plans, scientific reports, and technical summaries that support collaborative research execution and publication.
- Independently developed and maintain the Human Imprintome Catalog (humanicr.org), a public database-backed resource deployed on a DigitalOcean Linux server.

Research Scholar
North Carolina State University | Raleigh, NC
September 2015 -- October 2023

- Developed scalable, reproducible pipelines for genome-wide DNA methylation and RNA-seq analysis, enabling integration of large-scale multi-omics datasets.
- Led end-to-end bioinformatics analyses across toxicology, cancer genomics, immunology, and environmental health studies using high-throughput sequencing and complementary molecular data.
- Supported multi-institutional collaborations and large cohort projects through workflow development, quality-focused analysis, and integrative interpretation of complex omics datasets.

- Contributed to grant proposals, study protocols, and data analysis plans, strengthening the bioinformatics design of funded translational research.
- Refined and applied computational workflows for RNA-seq, DNA methylation, and other high-dimensional datasets to improve scalability, consistency, and biological interpretability.
- Collaborated closely with interdisciplinary investigators to translate analytical results into publications, presentations, and actionable scientific conclusions.

CNTS, Senior Bioinformatician

Walter Reed Army Institute of Research | Silver Spring, MD

June 2014 -- September 2015

- Conducted bioinformatics research in infectious disease, data integration, and application development for pathogen-focused sequencing studies.
- Redesigned a microbial pathogen discovery workflow and initiated a 16S metagenomics pipeline to improve analytical capability for surveillance-oriented research.
- Supported surveillance- and vaccine-related research through bioinformatics analysis, data integration, and scientific interpretation.
- Contributed genome assembly and downstream sequence analysis for *Candidatus Rickettsia asemboensis* strain NMRCii.
- Held U.S. Government Public Trust security clearance.

Bioinformatician I

Duke University Medical Center | Durham, NC

March 2007 -- June 2014

- Contributed to cancer genomics and B-cell biology research through computational analysis, scientific reporting, and manuscript development.
- Designed and implemented analytical frameworks for sequencing, gene expression, and epigenetic microarray studies.
- Analyzed biological data generated using diverse high-throughput platforms and translated results into publication-ready scientific outputs.
- Built, configured, and administered a Linux compute cluster with 16 Supermicro servers to support high-throughput bioinformatics workloads.

Research Associate

North Carolina State University | Raleigh, NC

September 2005 -- March 2007

- Conducted innate immunity research focused on natural killer cells.
- Performed independent research, expression microarray analysis, and molecular biology wet-lab work.
- Managed laboratory operations, equipment, and regulated biological materials.

Visiting Research Scholar

Oklahoma State University | Stillwater, OK

March 2005 -- September 2005

- Worked on database integration, parsing, Perl scripting, and MySQL-based data mining.

Assistant Lecturer

Ambo University | Ambo, Ethiopia

December 2000 -- August 2003

- Taught university-level courses and conducted independent academic research.

TECHNICAL PROFILE

Programming and Scripting: R/Bioconductor, Python, shell/Bash, awk, sed, SQL, NoSQL, JSON; prior experience with Perl

Platforms, Data, and Web: Linux/HPC, Nextflow, Git, PostgreSQL, MySQL, SQLite, full-stack scientific web development, HTML/CSS, Bootstrap, WordPress, DigitalOcean deployment, Linux server administration

Data Modalities: RNA-seq, Methyl-seq, ChIP-seq, FAIRE-seq, methylation arrays, exome sequencing, whole-genome sequencing, WGS, WES, Tempo-seq

AI and Computational Methods: Generative AI, large language models (LLMs), retrieval-augmented generation (RAG), Google Colab, AI-assisted coding and workflow development using ChatGPT and Codex

HONORS AND AWARDS

- **PRIDE OF THE WOLFPACK**, recognition for contributions to North Carolina State University, March 2026.
- **CHHE Outstanding Member Award**, North Carolina State University, 2019.
- **Netherlands Fellowship Program**, scholarship support for MS study.
- **Degree with Distinction**, Haramaya University.

PROFESSIONAL DEVELOPMENT

- Earned certificates for three multi-hour, instructor-led NCDS courses at UNC-Chapel Hill in 2026, completed on three separate days:
  1. Introduction to Generative AI for Coders (Accelerated).
  2. Advanced Generative AI for Coders: Foundations of AI.
  3. Deep Dive into Retrieval-Augmented Generation (RAG).
- Participated in a week of Seqera-hosted Nextflow professional development in Boston, MA, in 2026:
  1. Completed two days of Nextflow Advanced Training.
  2. Attended the two-day Nextflow Summit 2026.
- Participated in a one-day large language model (LLM) symposium at North Carolina State University in 2025, completing hands-on LLM and machine-learning exercises in Google Colab.

PROFESSIONAL AFFILIATIONS AND SERVICE

- Member, Center for Human Health and the Environment (CHHE), North Carolina State University.
- Member, Bioinformatics Research Center (BRC), North Carolina State University.
- Affiliate Member, University of North Carolina at Chapel Hill, Chapel Hill, NC, 2016--Present.
- Member, International Society of Developmental and Comparative Immunology, 2005--2007.
- Former editorial board member, Austin Medical Sciences Journal.
- Peer reviewer for genomics and biomedical journals, 3+ manuscripts annually.

KEY STRENGTHS

Scientific Expertise: Bioinformatics, computational genomics, epigenomics, DNA methylation, multi-omics, statistical genomics, study design, biological interpretation

Research Delivery: Pipeline development, reproducible workflows, high-throughput data analysis, cross-functional collaboration, scientific reporting

SELECTED PUBLICATIONS

- D. D. Jima, D. A. Skaar, A. Planchart, A. Motsinger-Reif, S. E. Cevik, and S. S. Park, "Genomic map of candidate human imprint control regions: The imprintome," *Epigenetics*, vol. 17, no. 13, pp. 1920–1943, 2022.
- J. Zhang et al., "The genomic landscape of mantle cell lymphoma is related to the epigenetically determined chromatin state of normal b cells," *Blood*, vol. 123, no. 19, pp. 2988–2996, 2014.
- J. Zhang, V. Grubor, C. L. Love, A. Banerjee, K. L. Richards, and P. A. Mieczkowski, "Genetic heterogeneity of diffuse large b-cell lymphoma," *Proceedings of the National Academy of Sciences*, vol. 110, no. 4, pp. 1398–1403, 2013.
- C. Love et al., "The genetic landscape of mutations in burkitt lymphoma," *Nature Genetics*, vol. 44, no. 12, pp. 1321–1325, 2012.
- D. D. Jima, J. Zhang, C. Jacobs, K. L. Richards, C. H. Dunphy, and W. W. L. Choi, "Deep sequencing of the small rna transcriptome of normal and malignant human b cells identifies hundreds of novel micrornas," *Blood*, vol. 116, no. 23, e118–e127, 2010.
- J. Zhang et al., "Patterns of microrna expression characterize stages of human b-cell differentiation," *Blood*, vol. 113, no. 19, pp. 4586–4594, 2009.

LINKS TO FULL PUBLICATIONS

PubMed: Search Dereje Jima

Google Scholar: Dereje Jima